

Western New England University
College of Engineering
ECE Department
Wave Transmission and Reception
EE 457/557
Fall 2025
Design Project #3
Due: November 17, 2025

Name: _____

References: _____

Design Project #3

	Score	Max
Design		200
HFSS		300
Tables		100
Presentation		100
Total		700

1. Design a 3-GHz coaxial-fed microstrip patch antenna. The substrate is 62-mil thick ($h = 1.5748$ mm) FR4 ($\epsilon_r = 4.4$ and $\tan \delta = 0.02$). The width of the patch, W , should be chosen as 1.5 times the electrical length of the patch ($W \approx 1.5L_e$).
2. Simulate the 3-GHz coaxial-fed microstrip patch antenna using HFSS. The patch and the ground plane are perfect electric conductors (pec) and have a thickness of $18 \mu\text{m}$ ($t = 18 \mu\text{m}$). In addition, assume that the length and width of the substrate are given by $L_s \approx 1.5L$ and $W_s \approx 1.5W$, respectively. Also, assume that the coaxial transmission line is an SMA connector. An SMA connector employs Teflon ($\epsilon_r = 2.1$) as the dielectric, an inner radius of 0.635 mm ($r_i = 0.635$ mm), and an outer radius of 2.032 mm ($r_o = 2.032$ mm). Next, assume that the thickness of the coaxial outer conductor is given by $r_o + r_i$ and the end cap's length is 1 mm. Lastly, assume that the maximum number of passes is 35 and the maximum delta S is 0.001 .
 - Tune the patch until the input return loss at 3 GHz is better than 30 dB ($RL_{in} \geq +30$ dB or $|\Gamma_{in}| \leq |S_{11}| \leq -30$ dB).
 - Determine a value for the directivity (D) and the gain (G) at a frequency of 3 GHz.
 - Determine a value for the E-Plane beamwidth (BW_{EP}) at a frequency of 3 GHz.
 - Determine a value for the H-Plane beamwidth (BW_{HP}) at a frequency of 3 GHz.
 - Determine a value for the radiation efficiency (e_{rad}) at a frequency of 3 GHz.
 - Plot $|S_{11}|$ over the range of -40 dB to 0 dB. Employ a frequency range of 2.8 GHz to 3.2 GHz.
 - Determine the bandwidth and percent bandwidth for which the input return loss is better than 10 dB ($RL_{in} \geq +10$ dB or $|\Gamma_{in}| \leq |S_{11}| \leq -10$ dB).
3. Complete the following tables (Table 1, Table 2, Table 3, and Table 4).

Table 1 Summary of the calculated design parameters for the 3-GHz coaxial-fed microstrip patch antenna.

Parameter	Calculated	
f_0	3	GHz
h	1.5748	mm
ϵ_r	4.4	
λ_g		mm
L_e		mm
W'		mm
ϵ_r^{eff}		
ΔL		μm
L_p		mm
W		mm
R_{Edge}		Ω
x_f		mm

Table 2 Summary of the simulated design parameters for the 3-GHz coaxial-fed microstrip patch antenna.

Parameter	V1	V2	V3	V4	
L_e					mm
ΔL					μm
L					mm
W					mm
x_f					mm
x_0					mm
L_s					mm
W_s					mm

Table 3 Summary of the simulated directivity and gain (at 3 GHz) for the 3-GHz coaxial-fed microstrip patch antenna.

Parameter	HFSS	
D		W/W
D		dB
G		W/W
G		dB
BW_{EP}		°
BW_{HP}		°
e_{rad}		%

Table 4 Summary of the calculated and simulated frequency response for the 3-GHz coaxial-fed microstrip patch antenna.

Parameter	Calculated	HFSS	
BW_f			MHz
BW_f			%